Multimodal Judges Flag, Humans Decide: Supplementary Material

A. The Decide Board

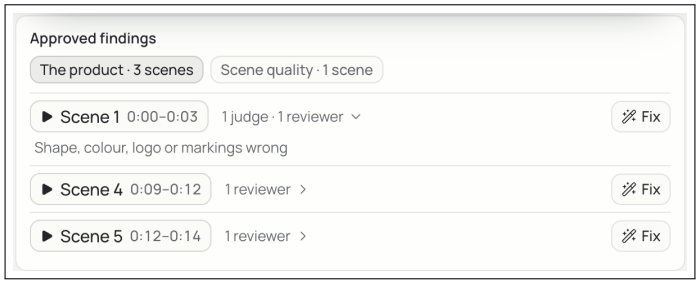


Figure S1. The decide board (cropped): an ad's approved findings by axis and scene, with how many judges and reviewers backed each and a Fix button that opens the matching edit tool on that scene.

B. The Same Ads with Judges Hidden

One reviewer who had not seen the ads reviewed the round's ads in a separate run of the portal with every judge hidden. The run held 60 of the 120 ads, three per product, in an order that cycles through the 20 products; every question was required and the secondary options were offered (sound was an optional question the portal could not require, so a blank answer counts as no issue; it was left blank on every ad). The units, the rules that turn answers into marks (those of the round), the outcomes, the bootstrap and the exclusions were written down and the analysis code hashed before the first review; one amendment, made before any review was submitted, records the scene-quality confound below. The reviewer submitted 60 of the 60 ads (all three ads of 20 products), and the results use only those ads. Five unsubmitted reviews under two other accounts were excluded. Checks: every judge was hidden, the run holds no thumbs, and the stored findings on the 60 ads are those of the round (360 of 360 ad–judge pairs unchanged). Pronunciation was computed but is not reported, because the narration engine and its judge changed after the round.

Table S1 compares, on the same ads and units, each reviewer alone, the stored judge findings and the round's reviewers, who saw them. A unit is marked as in the round: a scene with the product marked wrong or a swap the reviewer chose, a line marked bad, a caption marked for placement or relevance. Intervals are percentiles of 2,000 bootstrap resamples of ads. †In the round, marking a scene's product wrong pre-filled a swap for it, and an older grid hid such scenes from the scene-quality question, so the round was asked about fewer scenes (in parentheses) and its scene-quality marks mix the reviewer, the findings and the pre-fill.

Time is descriptive, since the groups are different people. Per-question gaps over 15 minutes count as breaks and are dropped; per-review times run from the first answer to submission, dropping reviews over 4 hours (one of the first reviewer's). The first reviewer took a median of 3.4 [2.4, 4.7] minutes from first answer to submission (quartiles in brackets; 59 reviews), against 4.1 [2.8, 6.6] for the round's 139 reviews of the same ads. Median time per question, first reviewer against the round: scene quality 44 s against 0 s; script 43 s against 65 s; caption placement 30 s against 7 s; caption relevance 1 s against 2 s. The product is answered first, and the review record starts at the first answer, so its time cannot be measured; in the round the product answer pre-filled the scene-quality grid, so its scene-quality time is near zero.

A second reviewer later reviewed all 60 ads in the same run with every judge hidden, under a second amendment written before any of these reviews was read. This reviewer had also reviewed the same 60 ads in the round about two weeks earlier, with judge findings on screen, and the amendment pre-specified the comparison of the two passes. On the same scenes, the product was marked wrong on 57% [49, 65] with judge findings and 19% [13, 26] with judges hidden (κ 0.22 between the passes); of the 7 lines the script judge flagged, 7 were marked with its findings on screen and 1 without. The pass with judges hidden came second, so memory of the findings would, if anything, narrow these gaps. A review by the second reviewer took a median of 2.3 [1.5, 4.4] minutes with judges hidden (59 timed reviews; quartiles in brackets), against 3.0 [2.3, 4.2] with judge findings in the round; this was the reviewer's second look at these ads, so the shorter time cannot be credited to hiding the judges. The two reviewers working alone agreed, on the 60 ads both reviewed, at Cohen's κ product 0.17 [0.02, 0.31]; scene quality 0.14 [0.04, 0.24]; script -0.04 [-0.05, -0.02]; caption placement 0.06 [-0.11, 0.25]; caption relevance -0.02 [-0.03, 0.00].

Over the session. Split into the first, middle and last 20 ads by submission time, each part holding 6 to 8 ads of each block, the second reviewer's product marks with judges hidden fell from 25% to 21% to 12% of scenes and the median review from 4.4 to 1.9 to 1.5 minutes, while on the same ads, with judge findings, this reviewer marked 61%, 50% and 59% and the round's majority 56%, 54% and 62%. Against the round's majority on each ad, the fall has Spearman ρ -0.27 over submission order (permutation p 0.04); total marks per ad have ρ -0.56 and minutes per review -0.59 (both $p < 0.001$). The first reviewer's product marks held (31%, 37% and 30%; ρ -0.06, p 0.65) while reviews sped up from 4.3 to 2.5 minutes (ρ -0.45, $p < 0.001$). This analysis was added after the results were read.

Table S1. The same ads with judges hidden (60 ads each). Alone: share of units the reviewer marked with every judge hidden. Judge: share the stored findings flagged. Alone's marks flagged: of the units the reviewer marked, the share the judge had flagged. Flags marked alone: of the judge's flags, the share the reviewer marked. Both columns give counts when fewer than 20. Round any and majority: share marked by any round reviewer, or by more than half of those asked, with judge findings on screen. κ: Cohen's κ between the reviewer alone and the round's majority on units both were asked. Shares in %, brackets are 95% intervals. Upper rows: the first reviewer; lower rows: the second reviewer, who was also one of the round's reviewers on these ads (the only one on 20 of them), so the round columns include that reviewer's own review and their κ against the majority is not given.

Axis	Unit	Units	Alone	Judge	Alone's marks flagged	Flags marked alone	Round any	Round maj.	κ alone vs maj.
First reviewer (60 ads)									
Product	scene	280	33 [25, 41]	68	74 [61, 85]	36 [27, 44]	67	58	0.11 [−0.02, 0.22]
Scene quality†	scene	280 (229)	47 [38, 55]	25	27 [20, 33]	49 [37, 61]	37	33	0.05 [−0.07, 0.17]
Script	line	280	3 [1, 4]	3	0 of 7	0 of 7	15	5	−0.03 [−0.05, −0.02]
Caption placement	caption	255	13 [8, 19]	no judge			17	5	0.06 [−0.06, 0.21]
Caption relevance	caption	255	5 [2, 8]	no judge			6	2	−0.03 [−0.05, −0.01]
Second reviewer (60 ads)									
Product	scene	280	19 [13, 26]	68	93 [85, 98]	26 [18, 34]	67	58	
Scene quality†	scene	280 (229)	31 [26, 36]	25	41 [31, 51]	49 [37, 62]	37	33	
Script	line	280	5 [3, 8]	3	1 of 15	1 of 7	15	5	
Caption placement	caption	255	16 [10, 21]	no judge			17	5	
Caption relevance	caption	255	1 [0, 3]	no judge			6	2	

C. Baselines in Detail

Table S2 lists the primary score of every product-fidelity baseline of Tab. 4 on the 120 blind-labeled scenes (29 defective, 35 ads), with 95% intervals from the same ad-clustered bootstrap (2,000 resamples) and the paired gap to the deployed clip judge. Lower predicted quality or higher dissimilarity counts as more likely defective; every sign and primary score was fixed before scoring and no threshold was tuned. Raising the clip judge's own threshold trades recall for precision: the first cut at which at least 70% of its flags are defective keeps 21 flags holding 15 of the 29 defective scenes, and the first at 85% keeps 8 holding 7. DOVER [43] and VideoScore2 [16] never see the catalog photos, so they cannot tell a wrong product from the right one; they show that a standard quality score would pass such scenes (VideoScore2 gives 107 of the 120 clips its top visual-quality score). The open judges received the deployed prompt, photos and output schema; Qwen3.5-9B [33] ran in bf16 and Qwen3.6-27B [34] in FP8, with thinking on as the primary setting (sampled at the temperature their model cards recommend, one sample per scene; thinking off was decoded greedily). A repeat with another seed and server layout moved the 9B's AUC by up to 0.09, so single rows carry that much run-to-run noise beyond the interval shown. The best open setting, the 27B with thinking off on eight frames, stays 0.12 below the clip judge, with an interval that includes zero, the only such gap besides the deployed judge's own eight-frame run; its verdicts reach κ 0.35 (17 of 29 defects in 36 flags), against 0.43 (26 in 56) for the clip judge. With thinking on, the open models flag 69 to 99 of the 120 scenes (κ 0.05 to 0.15). The prompt was written for the closed judge, and no variant was tried for the open ones. The eight-frame run of the deployed judge used a model version that the earlier runs did not record, so its gap may include a model change. The starting-frame judge is a separate checker with its own prompt, and its model version was not recorded either, so its gap to the clip judge mixes the input with the prompt and possibly the model; the eight-frame run keeps the deployed prompt.

Table S2. Product-fidelity baselines on the 120 blind-labeled scenes. Gap: clip judge minus baseline, paired, with 95% interval; intervals are not corrected for multiple comparisons.

Score	Input	AUC	95% interval	Gap to clip judge
Clip judge (deployed)	clip, catalog photos	0.84	[0.73, 0.94]	
Starting-frame judge	starting frame, photos	0.69	[0.61, 0.78]	+0.15 [0.01, 0.28]
Clip judge on eight frames	8 frames, photos	0.80	[0.66, 0.92]	+0.04 [−0.05, 0.15]
DINOv2 similarity	3 frames, photos	0.51	[0.39, 0.65]	+0.34 [0.18, 0.48]
CLIP similarity	3 frames, photos	0.57	[0.41, 0.72]	+0.27 [0.09, 0.47]
OCR text mismatch (EasyOCR)	3 frames, photos	0.57	[0.49, 0.64]	+0.28 [0.14, 0.42]
OCR text mismatch (Tesseract)	3 frames, photos	0.58	[0.50, 0.64]	+0.27 [0.14, 0.39]
DOVER (overall)	clip, no photos	0.48	[0.34, 0.63]	+0.37 [0.19, 0.56]
VideoScore2 (mean of 3 scores)	clip, text prompt	0.57	[0.44, 0.69]	+0.28 [0.12, 0.44]
Qwen3.5-9B, thinking on	8 frames, photos	0.67	[0.49, 0.84]	+0.18 [0.04, 0.31]
Qwen3.5-9B, thinking on	clip, photos	0.69	[0.55, 0.82]	+0.15 [0.04, 0.26]
Qwen3.6-27B (FP8), thinking on	8 frames, photos	0.72	[0.59, 0.86]	+0.12 [0.03, 0.22]
Qwen3.6-27B (FP8), thinking on	clip, photos	0.67	[0.53, 0.81]	+0.18 [0.06, 0.29]
Qwen3.6-27B (FP8), thinking off	8 frames, photos	0.73	[0.60, 0.86]	+0.12 [−0.04, 0.26]
Qwen3.6-27B (FP8), thinking off	clip, photos	0.68	[0.57, 0.80]	+0.16 [0.01, 0.29]
Qwen3.5-9B, thinking off	8 frames, photos	0.54	[0.39, 0.71]	+0.30 [0.15, 0.46]
Qwen3.5-9B, thinking off	clip, photos	0.60	[0.48, 0.74]	+0.25 [0.10, 0.38]

D. Label Sets Behind the Numbers

Table S3 lists two more judge experiments of Sec. 6, with their labels and answers. Table S4 lists each label set behind the judge and review numbers, who made it, whether its makers could see any judge output, and which claims it supports; the fix tests' labels are in Tab. 3.

Table S3. Other judge experiments. Labels: **B** people blind to judge output, **M** a model or measurement, **L** one listener per line; numbers on labels other than **B** measure agreement with that reference only. Each choice was made and scored on the same labels, so numbers are optimistic; brackets are 95% intervals.

Axis and question	Labels	Answer and key numbers
Pronunciation. Which speech-to-text model; recheck on audio?	**L** 208 lines: audio only (blind), then with script and judge output	**Error-keeping, text-only recheck.** Newer speech-to-text models returned the script word for word on 30% and 46% of defect lines (ours 6%; labeled after output was shown); audio in the recheck cut recall from 44 to 38 of 54
Sound. Does the video model talk over the narration?	**M** 50 clips, another format; separated-stem loudness	**Yes; count words to find it.** A word count on the raw clip audio caught 9 of 9 with no false alarms, an audio-model judge 6 of 9

E. Judge Models

The judges whose findings reviewers saw in the round run on closed models: Gemini Flash for the product clip judge, the scene clip judge, the sound judge and the listing-photo text reader; GPT-5.6-sol for the scene frames judge; Azure Speech with GPT-5.6-luna for pronunciation; and Claude Sonnet 5 for the script judge. The caption judges, run only after the round, use Gemini Flash on the whole rendered ad.

F. Review Rubric

Table S5 lists the questions of the review rubric as they were used in the 120-ad round (Sec. 4). In that round the run made every question required, offered the secondary options and showed judge findings inside the questions. The scene-quality grid asked for one mark on every rateable scene, meaning a scene with a generated clip or an animated still and never the end card; for most of the round the grid skipped scenes already marked product wrong and counted them as swaps.

Table S4. Label sets. "Blind" means the labelers saw no judge output. Assisted labels were made with judge findings on screen, and a thumbs up wrote the finding into the answer.

Label set	Size	Labelers	Blind?	Used for
Annotator product labels on production clips	120 decided scenes of 146 labeled, 29 defective	four annotator accounts; two votes on most scenes, one on five; they miss subtle defects	yes	product judge and baselines (Tab. 4, Tab. S2)
Physics pilot	10 ads	same annotators	yes	scene judges, swaps and impossible motion (Tab. 2)
Earlier review round	66 accepted scene flags	reviewer thumbs	assisted	what scene judges flag
Scene judges, 120-ad round	489 reviewer-scene pairs	portal reviewers	assisted	one scene judge or two (Tab. 4)
Pronunciation corpus	208 narrated lines	five listeners, one per line, ten lines heard by all	first pass yes, second pass after reveal	speech-to-text model, recheck (Tab. S3), precision and recall
Fresh pronunciation sample	100 production lines (70 flagged, 20 near misses, 10 passed)	one listener	yes	precision on fresh lines (Tab. 2)
Talking-over clips	50 of 128 clips	stem-separation loudness	no people	sound (Tab. S3)
Script checker study	155 ads, 200 ratings	four raters, shown both checkers' calls	no	reading listing photos
Caption placement	100 captions	models against themselves	no people	placement pick stability (Tab. 4)
Caption marks, 120-ad round	530 captions	portal reviewers	yes, no caption judge shown	caption judges (Tabs. 2 and 4)
120-ad review round	562 scenes, 278 reviews	five internal reviewers	assisted	split, time, agreement (Sec. 4)
People alone	60 of the round's ads, 280 scenes	two reviewers, 60 ads each	first yes; the second had seen judge findings in the round	the same ads with judges hidden (Supp. B)

Table S5. Questions of the review rubric in the 120-ad round, in the order the portal shows them. The last answer of each question is the problem answer; it opens the picker and the secondary options.

Question	Answers	A problem answer points to	Secondary options
The product	Product looks right; Product wrong	every scene where it happens	Wrong product; Shape, colour, logo or markings wrong; AI issue; Product text not intact, graded Low ("still readable") or High ("gibberish"); a comment per scene
Scene quality	No opinion; Keep; Swap out	each rateable scene, which gets its own mark	AI issue; Repetitive, dull or crowded scene; World or continuity breaks; Something moves impossibly; Weak ending (last scene only); free-text boxes for another reason and for why
Pronunciation	Sounds right; Mispronounced	every spoken line where it happens	none; an optional box for the word
Sound quality	Sounds right; Issues	the whole ad	"Which part?", at least one of The voice and The music; an optional note
Script quality	Good; Bad	every spoken line where it happens	Factually incorrect (an unsupported claim); Awkward or unnatural phrasing; an optional note
Overlay text: placement	Correct; Can be better	every captioned scene where it happens	none
Overlay text: relevance	Correct; Issues	every captioned scene where it happens	none